

EXP-3:DECISION TREE ALGORITHM:

PROGRAM:

```
import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier


# Sample Dataset

data = {

    "Attendance": ["High", "High", "Medium", "Low", "Low", "Medium", "High", "Low"],

    "Assignment": ["Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No"],

    "StudyHours": ["High", "Medium", "High", "Low", "Medium", "High", "Medium", "Low"],

    "InternalMarks": ["High", "Medium", "High", "Low", "Medium", "Medium", "High", "Low"],

    "Result": ["Pass", "Pass", "Pass", "Fail", "Fail", "Pass", "Pass", "Fail"]

}


df = pd.DataFrame(data)


print("Dataset:")

print(df)


# Encode categorical data

encoders = {}

for col in df.columns:

    le = LabelEncoder()

    df[col] = le.fit_transform(df[col])

    encoders[col] = le


# Features and Target

X = df.drop("Result", axis=1)

y = df["Result"]
```

```
# Train ID3 Decision Tree
```

```
model = DecisionTreeClassifier(criterion="entropy")
```

```
model.fit(X, y)
```

```
# New Sample
```

```
new_sample = pd.DataFrame({
```

```
    "Attendance": [encoders["Attendance"].transform(["Medium"])[0]],
```

```
    "Assignment": [encoders["Assignment"].transform(["Yes"])[0]],
```

```
    "StudyHours": [encoders["StudyHours"].transform(["High"])[0]],
```

```
    "InternalMarks": [encoders["InternalMarks"].transform(["Medium"])[0]]
```

```
})
```

```
# Prediction
```

```
prediction = model.predict(new_sample)
```

```
print("\nNew Sample:")
```

```
print({
```

```
    "Attendance": "Medium",
```

```
    "Assignment": "Yes",
```

```
    "StudyHours": "High",
```

```
    "InternalMarks": "Medium"
```

```
})
```

```
print("Predicted Result:", encoders["Result"].inverse_transform(prediction)[0])
```

```
# Decision Tree Structure
```

```
print("\nDecision Tree Structure")
```

```
print("-----")
```

```
print("""
```

```
    Attendance
```

```
    /    \
```

High	Medium/Low		
Pass	StudyHours		
/	\\		
High	Low		
Assignment	InternalMarks		
/	\\	/	\\
Yes	No	High	Low
Pass	Fail	Pass	Fail

""")

OUTPUT:

```

*** Dataset:
  Attendance Assignment StudyHours InternalMarks Result
0      High      Yes      High      High      Pass
1      High      No      Medium     Medium     Pass
2     Medium     Yes      High      High      Pass
3       Low      No       Low       Low      Fail
4       Low     Yes     Medium     Medium     Fail
5     Medium     No      High      Medium     Pass
6      High     Yes     Medium     High      Pass
7       Low      No       Low       Low      Fail

New Sample:
{'Attendance': 'Medium', 'Assignment': 'Yes', 'StudyHours': 'High', 'InternalMarks': 'Medium'}
Predicted Result: Fail

Decision Tree Structure
-----
      Attendance
     /         \
   High       Medium/Low
   |           |
   Pass      StudyHours
             /      \
          High      Low
          |          |
        Assignment InternalMarks
        /  \        /  \
      Yes No High Low
      |  |  |  |  |
      Pass Fail Pass Fail
  
```